

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
Academic Year	2025–2026 (Even)	Batch	2023–2027
Name	Kishor Sreenivasan	Register No.	3122235001068
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Experiment 4: Binary Classification using Linear and Kernel-Based Models

1. Aim and Objective

To classify emails as spam or ham using Logistic Regression and Support Vector Machine (SVM) classifiers and to analyze the effect of hyperparameter tuning on classification performance.

2. Dataset Description

The Spambase dataset is a benchmark binary classification dataset used to identify spam emails. It consists of numerical features extracted from email content and a binary class label indicating spam or non-spam.

Dataset Reference:

- Kaggle – Spambase Dataset

3. Preprocessing Steps

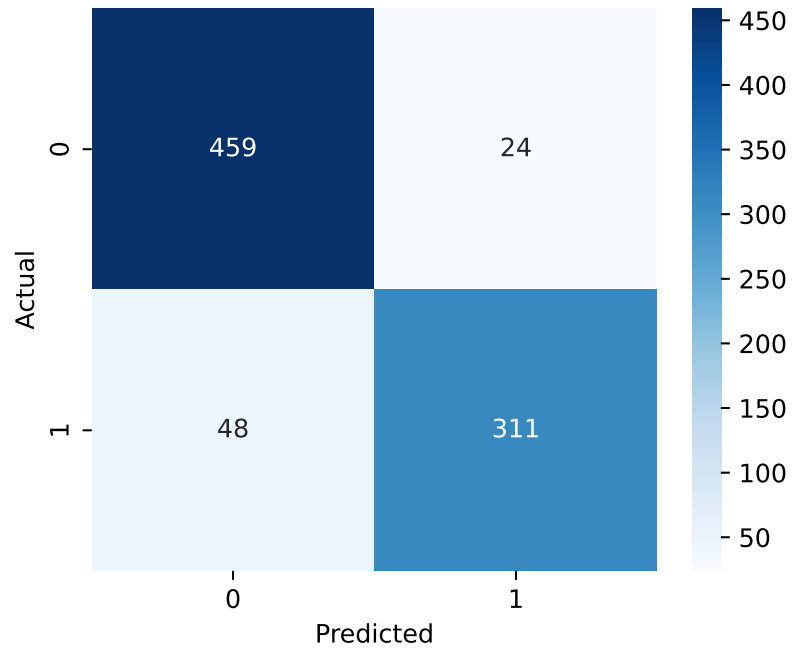
- Loaded the dataset using Pandas
- Separated features and target labels
- Performed stratified train-test split
- Applied feature scaling using StandardScaler

4. Implementation Details

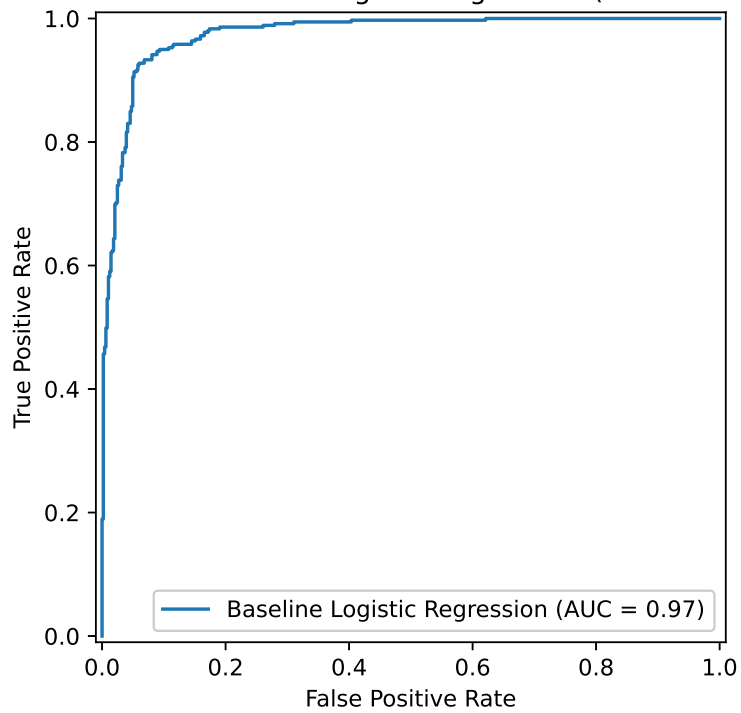
- Implemented Logistic Regression, baseline and CV
- Tune Logistic Regression hyperparameters.
- Train SVM with different kernels.
- Tune SVM hyperparameters.
- Evaluate models using standard metrics.
- Perform 5-Fold Cross-Validation.

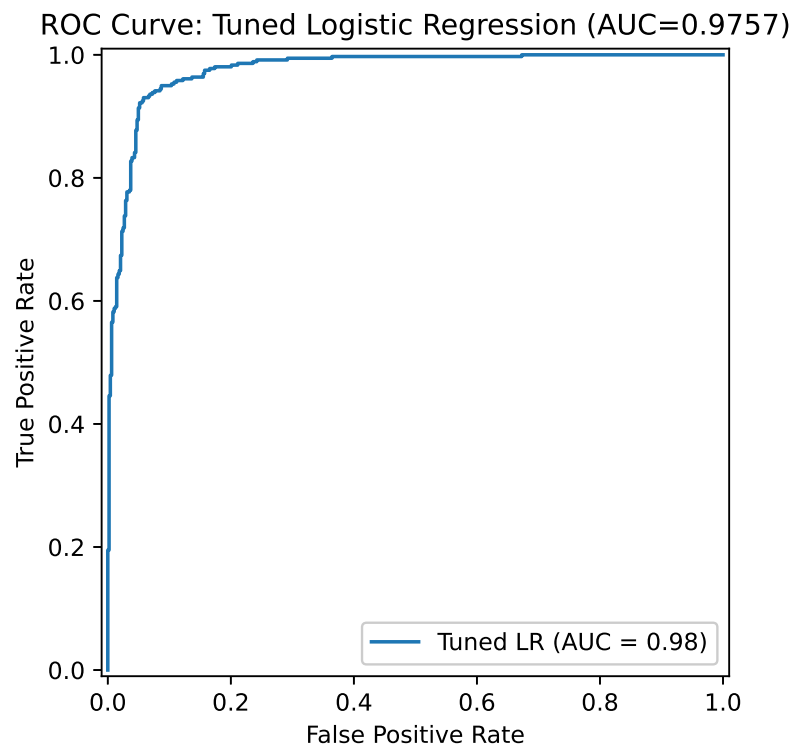
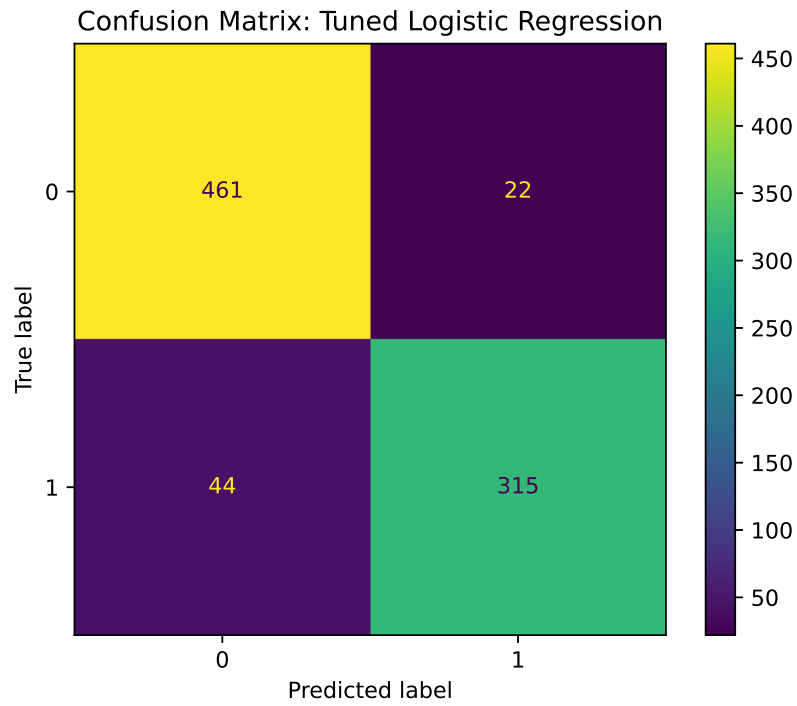
5. Visualisations

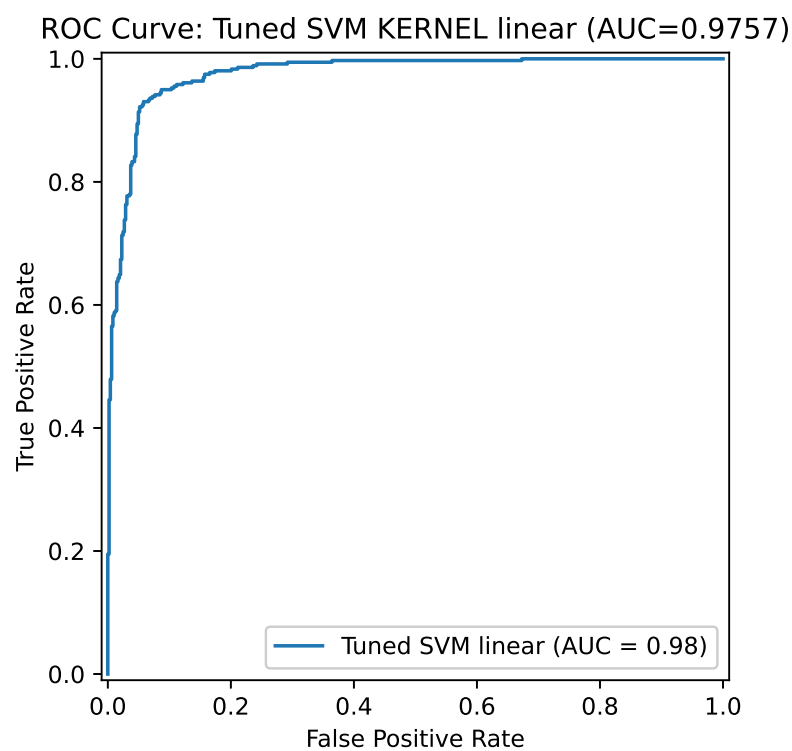
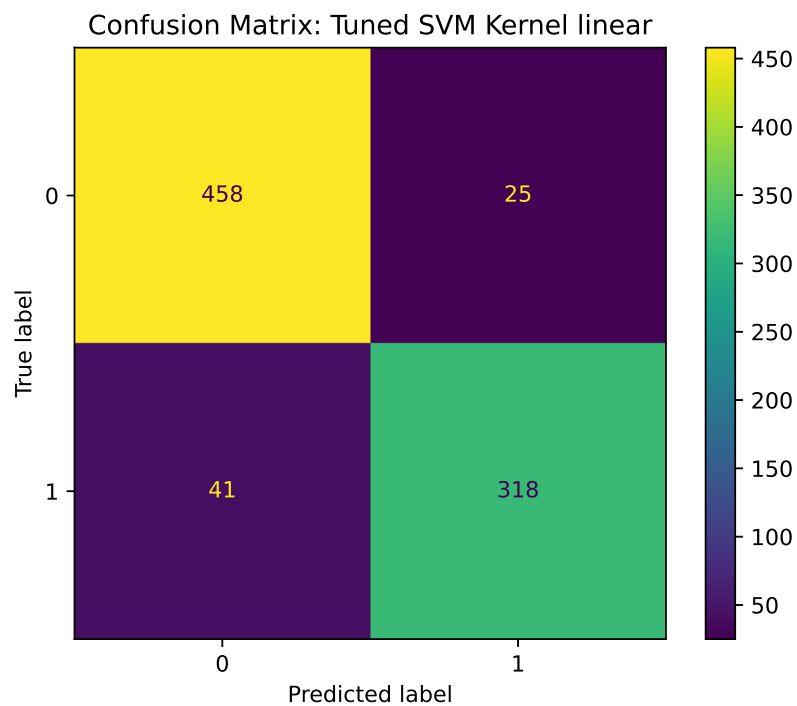
Confusion Matrix: Baseline Logistic Regression

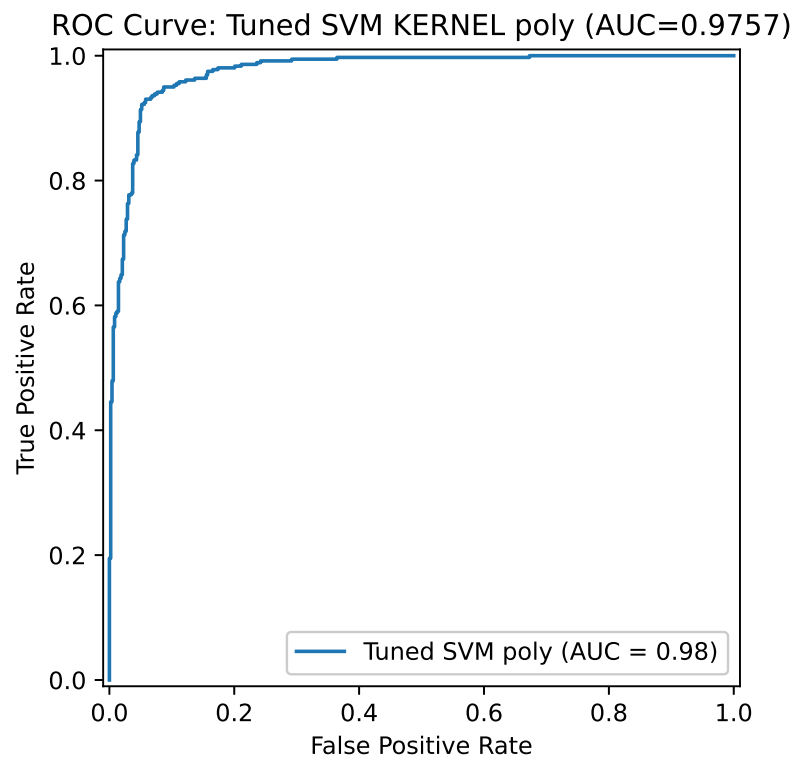
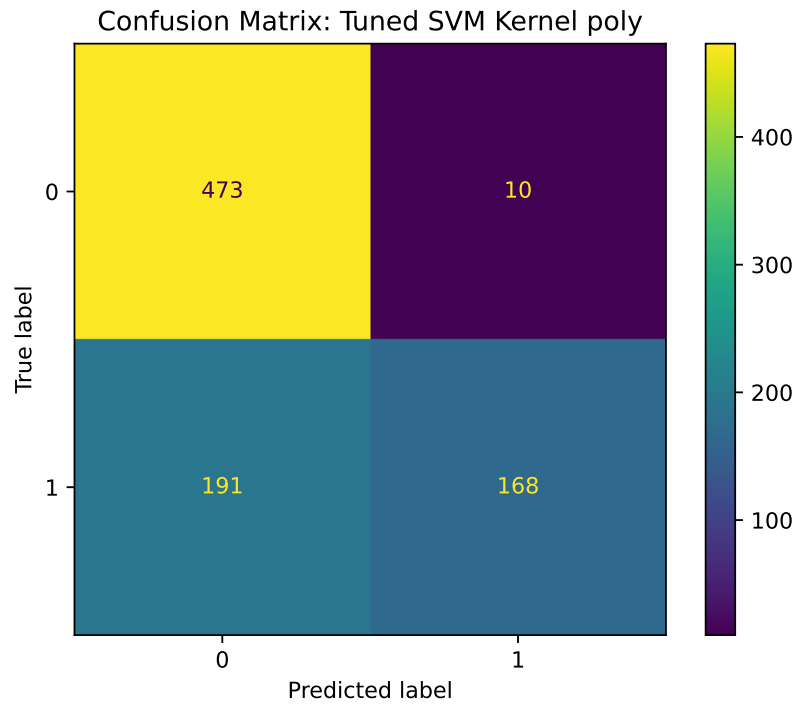


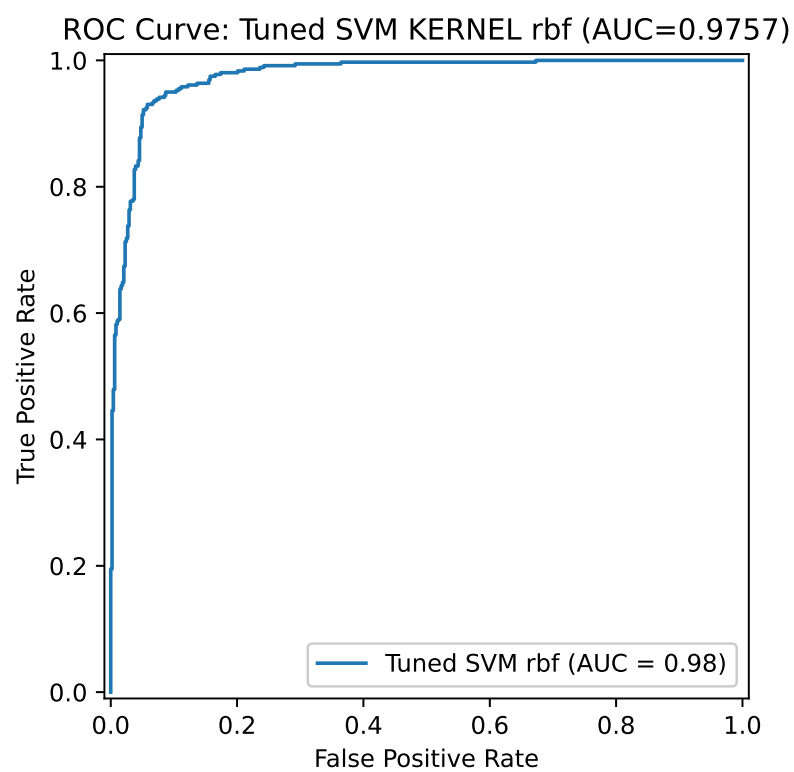
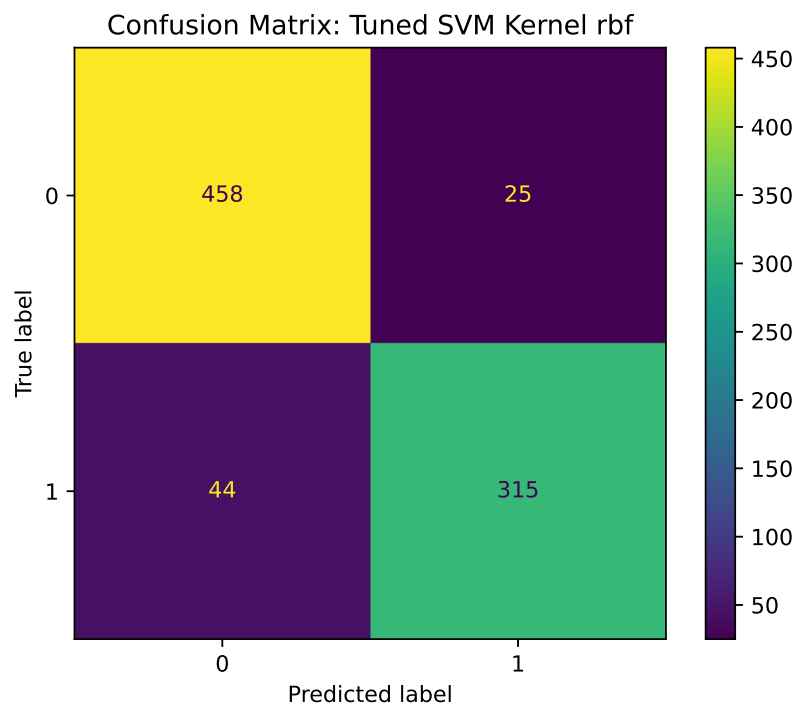
ROC Curve: Baseline Logistic Regression (AUC=0.9747)

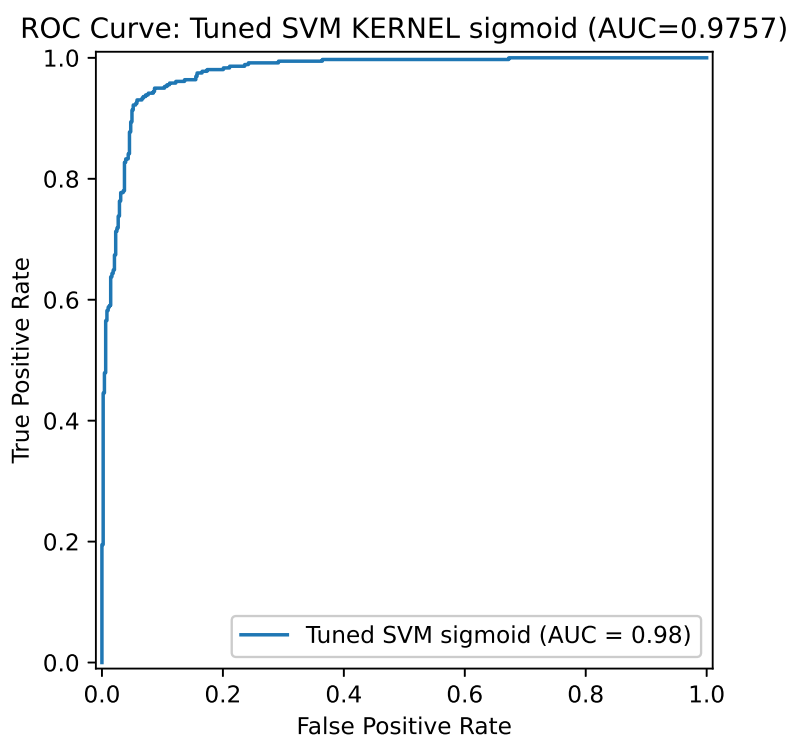
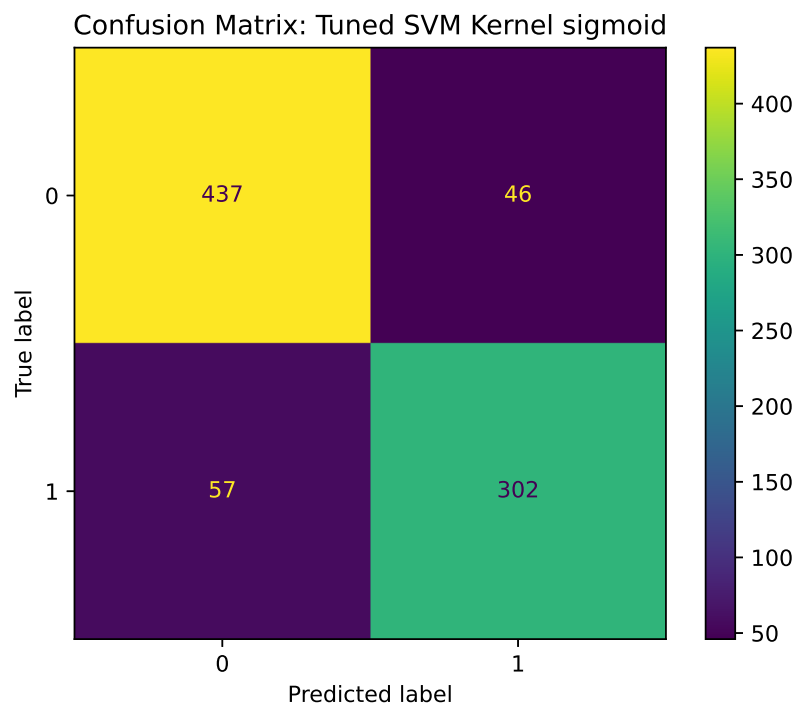


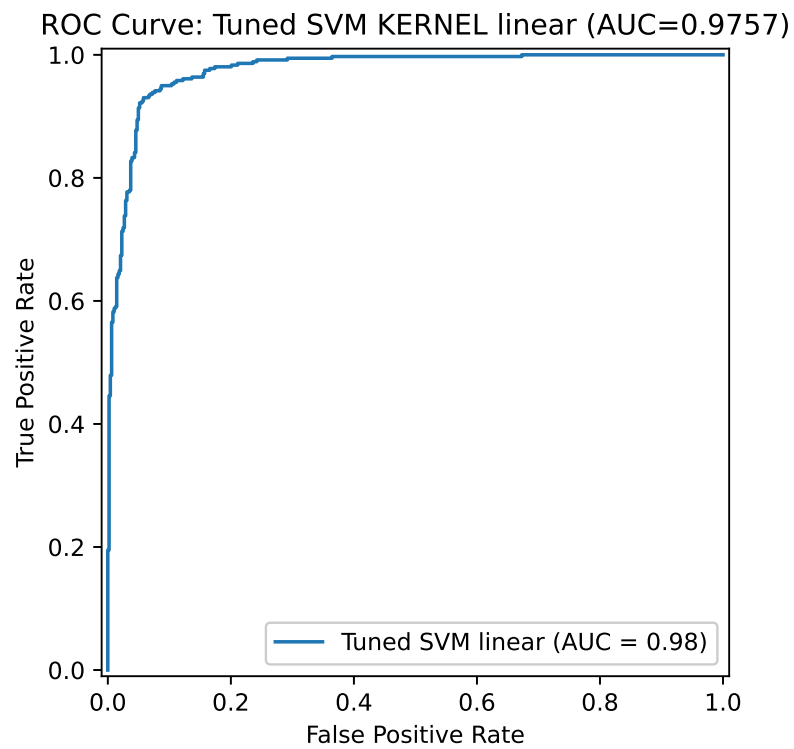
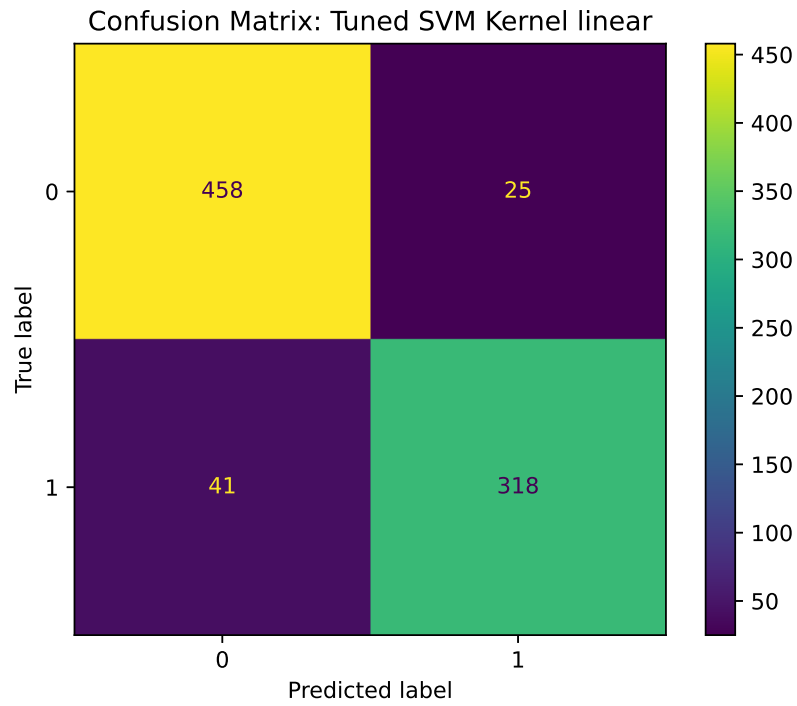


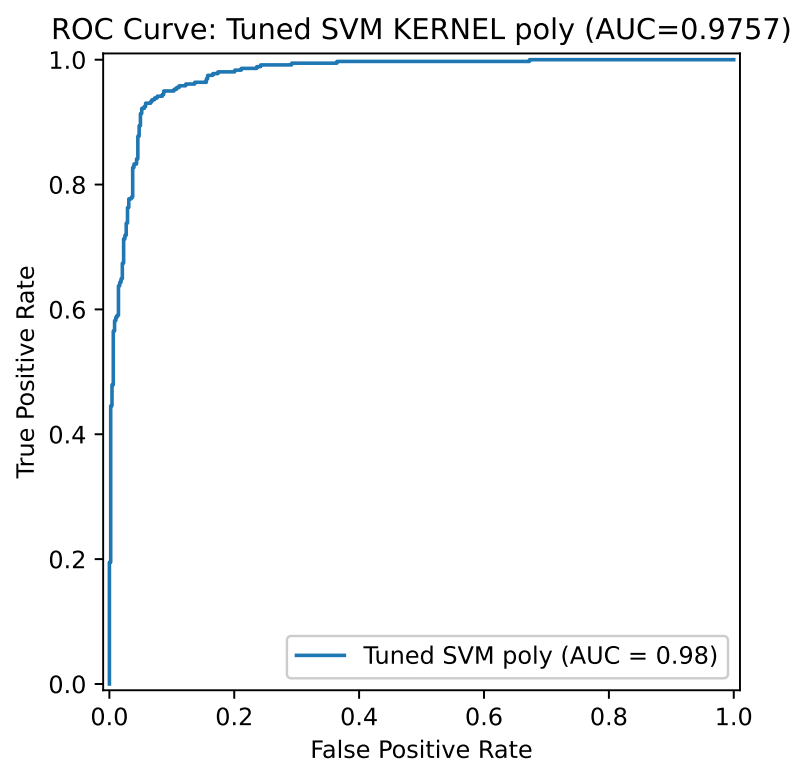
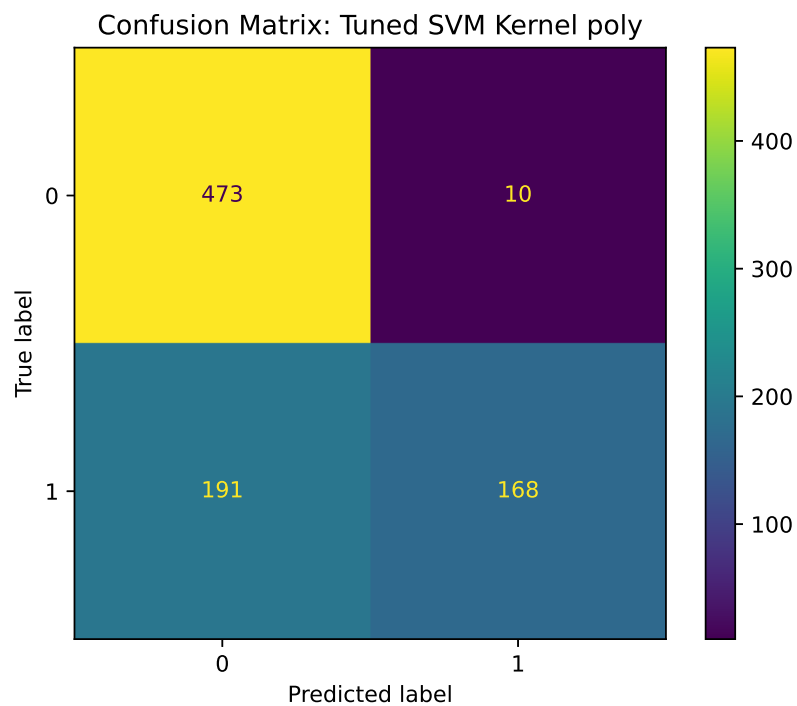


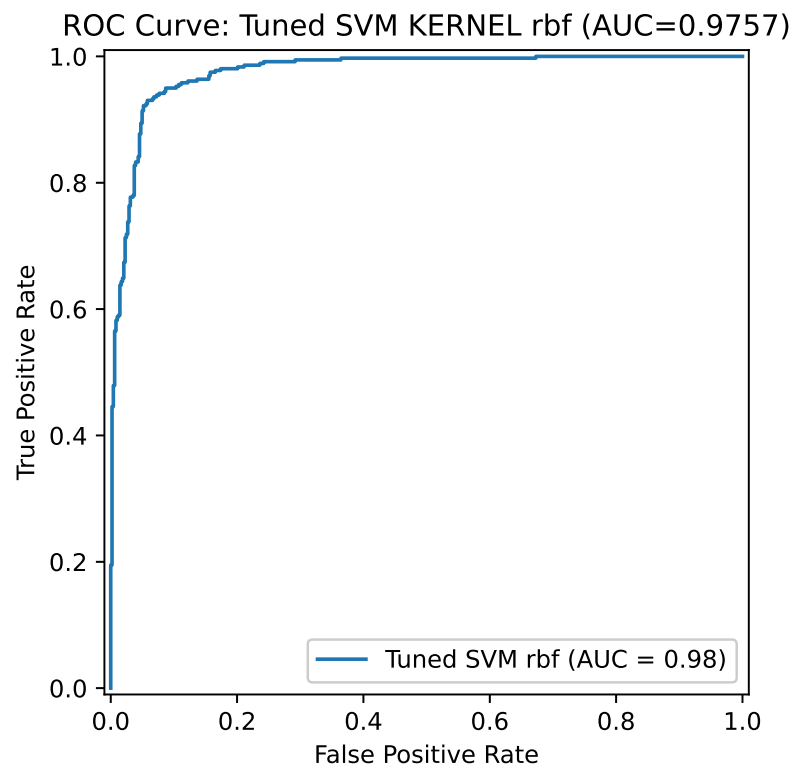
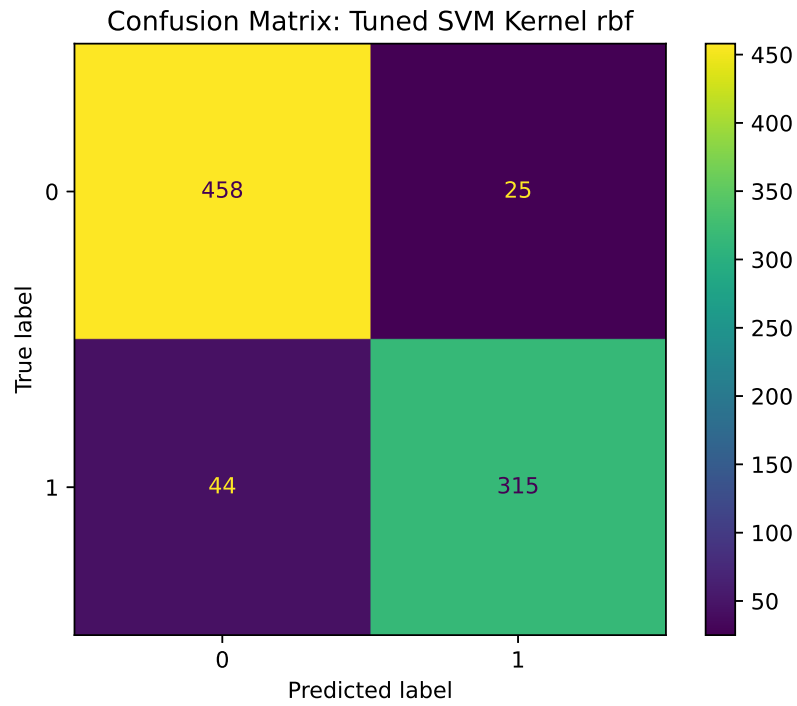


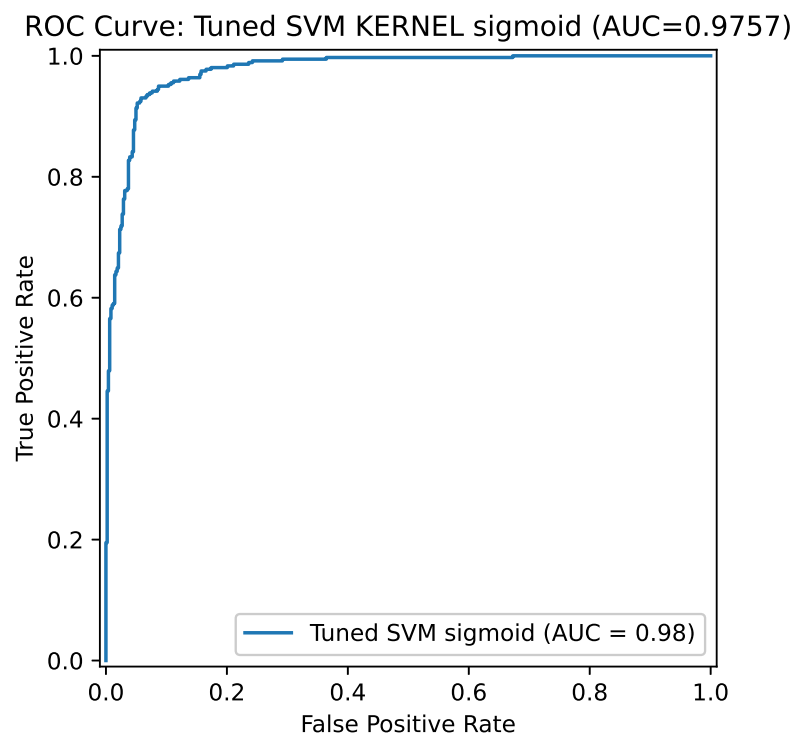
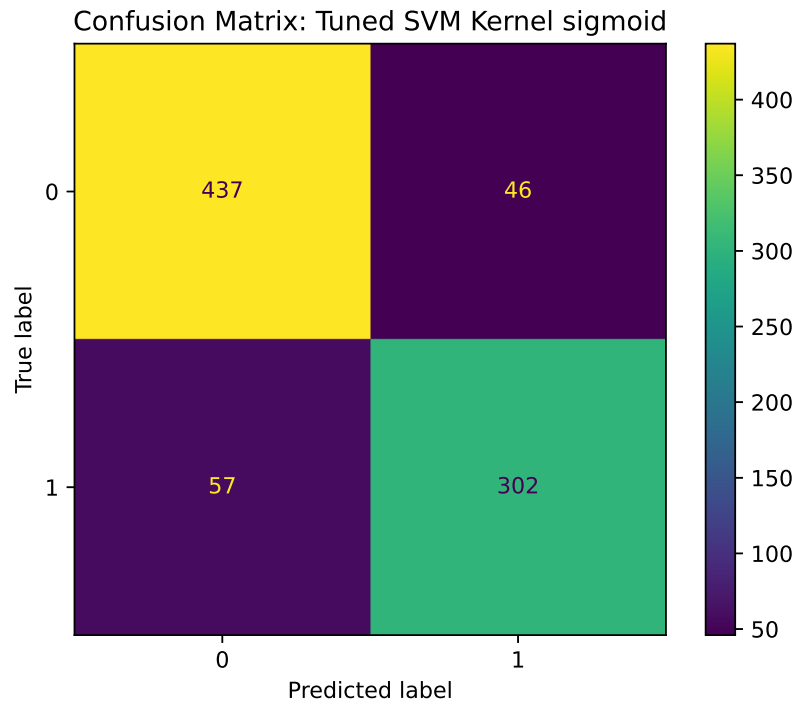


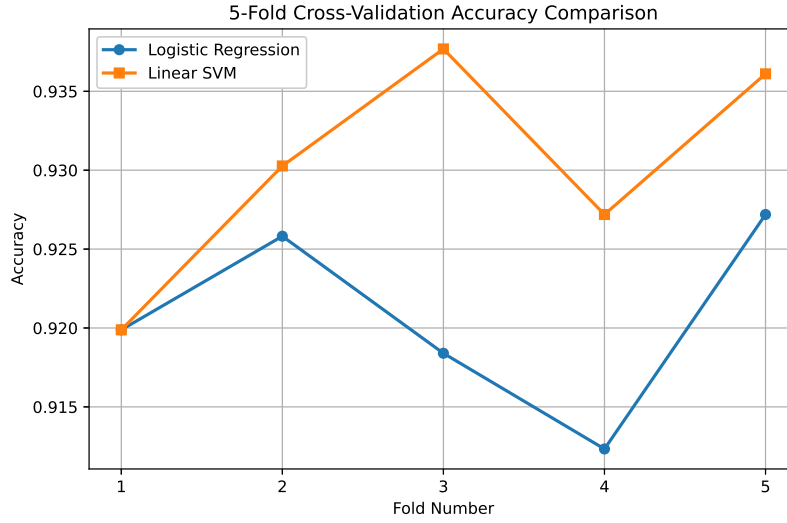












6. Performance Tables

Model	Search Method	Best Parameters	Best CV Accuracy
Logistic Regression	Grid / Random	c:10, penalty: l2, liblinear	0.9207
SVM	Grid / Random	c:1, gamma:scale, kernel: rbf	0.9302

Table 1: Hyperparameter Tuning Results

Metric	Value
Accuracy	0.9216
Precision	0.9347
Recall	0.8777
F1 Score	0.9052
Training Time (s)	0.8024s

Table 2: Logistic Regression Performance

Kernel	Accuracy	F1 Score	Training Time (s)
Linear	0.9216	0.9060	1.1670
Polynomial	0.7613	0.6257	1.0201
RBF	0.9131	0.9013	0.6066
Sigmoid	0.8777	0.8543	0.5147

Table 3: SVM Kernel-wise Performance

Fold	Logistic Regression	SVM
Fold 1	0.9199	0.9199
Fold 2	0.9258	0.9303
Fold 3	0.9184	0.9327
Fold 4	0.9123	0.9272
Fold 5	0.9272	0.9361
Average	0.9207	0.9302

Table 4: K-Fold Cross-Validation Results (K = 5)

Criterion	Logistic Regression	SVM
Accuracy	0.9207	0.9302
Model Complexity	Low	High
Training Time	0.8024	0.6066
Interpretability	High	Low

Table 5: Comparative Analysis

Performance Analysis and Discussion

1. Overall Model Comparison

In this experiment, the **Support Vector Machine (SVM)** using a RBF kernel emerged as the top performer.

- **SVM (Linear Kernel):** Achieved the highest test accuracy of 0.9216 and a peak F1 score of 0.9060, though in execution, RBF is faster.
- **RBF:** Followed closely with an accuracy of 0.9131 and an F1 score of 0.9013.
- **Insight:** Both margin-based optimization and probabilistic modeling are highly effective for the Spambase dataset.

2. The Role of Regularization

Regularization was pivotal in refining the Logistic Regression model.

- **Optimal Setup:** The grid search identified $L2$ (Ridge) regularization with an inverse strength of $C = 10$ using the `liblinear` solver.
- **Feature Selection:** $L2$ regularization enabled implicit feature selection by zeroing out less informative attributes.
- **Reliability:** A high Precision of 0.9347 indicates the model is excellent at minimizing "false positives," which is critical for preventing legitimate emails from being flagged as spam.

3. Impact of SVM Kernel Selection

The choice of kernel profoundly influenced the SVM's ability to map the data.

- **Linear Kernel:** While selected as "most stable" during tuning, it was slightly edged out by the RBF kernel in final testing.
- **RBF Kernel:** It is the superior kernel, with quick runtime and high accuracy and F1 score.
- **Polynomial Kernel:** Significantly underperformed with an accuracy of 0.7613, indicating ineffective feature mapping for this specific dataset.

4. Efficiency and Generalization

An assessment of the bias-variance trade-off reveals distinct computational profiles:

- **Speed:** Logistic Regression offered the lowest complexity and a fast training time.
- **Complexity:** The Linear SVM was more expensive, requiring 1.1670 seconds to train.
- **Generalization:** The consistency between Average CV scores (LR: 0.9207; SVM: 0.9302) and final test results confirms that both models generalize well to unseen data.

Learning Outcomes

- Understood classifiers built upon margin-based and probabilistic criterion.
- Applied hyperparameter tuning to improve model performance.
- Evaluated various models according to performance metrics.
- Interpreted experimental results to a thorough conclusion.

References

- https://scikit-learn.org/stable/modules/linear_model.html *logistic-regression* Scikit – learn : LogisticRegression
- <https://scikit-learn.org/stable/modules/svm.html> Scikit-learn: Support Vector Machines
- https://scikit-learn.org/stable/modules/grid_search.html *Scikit – learn : Hyperparameter Optimization*
- <https://www.kaggle.com/datasets/somesh24/spambase> Spambase Dataset – Kaggle
- <https://archive.ics.uci.edu/ml/datasets/spambase> UCI ML Repository – Spambase